

A Sharp Refinement of Serfling's Inequality at the Variance Scale

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Abstract

Sampling without replacement becomes increasingly concentrated as the sample size approaches the population size and becomes deterministic when the entire population is observed. Serfling's inequality captures part of this reduction in variability, but its correction is larger than the factor in the exact variance of the sample mean. We show that the factor from the exact variance remains valid in a quadratic moment generating function bound for every bounded population, and the result extends to arbitrary signed weighted statistics. We then determine the smallest constant, depending only on the population size N , for which the quadratic bound holds uniformly when only the range of the population is used. The constant has different forms for even and odd N . The proof reduces the weighted problem to centered hypergeometric moment generating functions and uses a recursion that lowers both the population size and the hypergeometric sample size while preserving the dependence on that sample size. The same analysis determines the smallest uniform constant for weighted contrasts of bounded exchangeable random variables.

1 Introduction

Sampling without replacement becomes increasingly concentrated as the sample size approaches the population size and becomes deterministic when the entire population is observed. This reduction in variability is already visible in the exact variance formula for the sample mean.

To make this precise, let $N \geq 2$ and let $X = (X_1, \dots, X_N) \in [a, b]^N$ be a fixed finite population. Let π be a uniformly random permutation of $[N]$. For $1 \leq n \leq N$, let $S_n = \{\pi(1), \dots, \pi(n)\}$ and write

$$\bar{X}_N = \frac{1}{N} \sum_{i=1}^N X_i, \quad \bar{X}_{S_n} = \frac{1}{n} \sum_{i=1}^n X_{\pi(i)}.$$

Writing $\sigma_N^2 = N^{-1} \sum_{j=1}^N (X_j - \bar{X}_N)^2$ for the finite-population variance, sampling without replacement gives

$$\text{Var}(\bar{X}_{S_n}) = \frac{\sigma_N^2}{n} \frac{N-n}{N-1}.$$

Thus $(N-n)/(N-1)$ is exactly the factor by which sampling without replacement reduces the variance relative to independent sampling with replacement.

Classical Hoeffding bounds do not reflect this full reduction in variability. Hoeffding's comparison theorem ([Hoeffding, 1963](#), Theorem 4) transfers exponential bounds from sampling with replacement

to sampling without replacement, but the resulting bound retains the dependence of the corresponding with-replacement bound and therefore does not capture the full finite-population reduction.

Serfling (1974) introduced a finite-population correction that partially reflects this effect. For $u > 0$, Serfling’s inequality gives

$$\mathbb{P}\{\sqrt{n}(\bar{X}_{S_n} - \bar{X}_N) \geq u\} \leq \exp\left\{-\frac{2u^2}{(1 - f_n^*)(b - a)^2}\right\}, \quad f_n^* = \frac{n - 1}{N}.$$

Serfling’s correction is $1 - f_n^* = (N - n + 1)/N$, whereas the exact sampling variance involves

$$1 - f_n = \frac{N - n}{N - 1}, \quad f_n = \frac{n - 1}{N - 1}. \quad (1)$$

The difference is most visible when almost the entire population is observed. If $n = N - 1$, Serfling’s correction is $2/N$, while the factor in the exact sampling variance is $1/(N - 1)$. Serfling suggested the latter as a possible refinement (Serfling, 1974, p. 47). Greene and Wellner (2016) later noted that a definitive answer was not known, and Greene and Wellner (2017, equation (1.4)) asked explicitly whether the replacement is valid for every bounded population, or even for binary populations.

This leaves a precise question. Can Serfling’s correction be replaced uniformly by the factor $(N - n)/(N - 1)$ appearing in the exact sampling variance?

We answer this question affirmatively, and the result extends beyond the sample mean. For arbitrary signed weighted statistics, the coefficient vector enters both the variance and a quadratic moment generating function (MGF) bound through the same quantity. The sample-mean correction is therefore a special case of a more general weighted result.

For equal weights, the general result reduces exactly to the finite-population correction $(N - n)/(N - 1)$ in (1). Hence Serfling’s correction can be replaced by the factor appearing in the exact sampling variance for every bounded population. We also determine the smallest constant, depending only on N , that makes the corresponding quadratic bound valid uniformly when only the range of the population is used. Its value depends on whether N is even or odd.

The weighted formulation also connects the finite-population problem to concentration for weighted exchangeable variables. Lee and Kim (2026) obtained an explicit constant of the form $1 + O(N^{-1})$ by reducing the weighted problem to centered hypergeometric MGFs. They also characterized the smallest possible uniform constant by optimizing over this hypergeometric family, but did not evaluate the optimization exactly. The present paper evaluates this remaining optimization. The key new ingredient is a recursion relating parameters (N, K, m) to $(N - 2, K - 1, m - 1)$ while preserving the factor that depends on the hypergeometric sample size. This yields the exact uniform constant and, for equal weights, the refinement of Serfling’s inequality.

1.1 Contributions

The paper makes three main contributions.

- We resolve the finite-population correction question by showing that $(N - n)/(N - 1)$ can replace Serfling’s correction $(N - n + 1)/N$ for every bounded population. The result follows from a more general inequality for arbitrary signed weighted statistics.
- We determine the smallest constant depending only on N for which the resulting quadratic MGF inequality holds uniformly over bounded populations, sample sizes, coefficient vectors,

and $t \in \mathbb{R}$. The smallest constant has different explicit forms for even and odd population sizes.

- We evaluate the centered hypergeometric optimization left open by [Lee and Kim \(2026\)](#). This determines the smallest uniform constant for weighted contrasts of bounded exchangeable random variables and shows that the finite-population and exchangeable problems lead to the same optimization over centered hypergeometric MGFs.

1.2 Related Work

Our work is most closely related to finite-population concentration, more general concentration under sampling without replacement, and concentration for weighted exchangeable sums.

Finite-population concentration and the Serfling problem. The study of exponential concentration under sampling without replacement goes back at least to [Hoeffding \(1963\)](#), whose comparison principle transfers exponential bounds from independent sampling to sampling without replacement. [Serfling \(1974\)](#) strengthened this comparison by introducing an explicit finite-population correction that reflects the reduction in variability as the sampling fraction increases. A related literature studies concentration specifically for the hypergeometric distribution. [Chvátal \(1979\)](#) obtained an early exponential tail bound, and [Hush and Scovel \(2005\)](#) later derived an improved hypergeometric concentration inequality. [Bardenet and Maillard \(2015\)](#) further refined Serfling-type bounds and developed Bernstein–Serfling and empirical Bernstein–Serfling inequalities. [Greene and Wellner \(2016\)](#) reviewed finite-sampling and hypergeometric inequalities in connection with two-sample empirical processes, while [Greene and Wellner \(2017\)](#) established exponential bounds for the hypergeometric distribution and a convex-order reduction under which binary populations are extremal for the relevant convex functionals. More recent work has obtained complementary tail bounds through majorization and direct hypergeometric arguments ([Ai et al., 2025](#); [Mannalath et al., 2026](#)). These results address several aspects of finite-population concentration, but they do not settle whether the factor in the exact sampling variance gives a uniform quadratic MGF bound.

General concentration under sampling without replacement. A broader literature studies concentration for general functions of random subsets and permutations. This includes approaches based on concentration on slices of the discrete cube, modified log-Sobolev inequalities on the multislice, and empirical-process methods ([Tolstikhin, 2017](#); [Sambale and Sinulis, 2022](#); [Polaczyk, 2023](#); [Yang, 2025](#)). These results apply to substantially more general functionals than the linear statistics considered here. Our question is narrower. We seek the smallest quadratic MGF constant that uses only the bounded range while retaining the exact dependence on the sampling fraction.

Weighted exchangeable sums. A closely related literature studies concentration under exchangeability. [Barber \(2024\)](#) established Hoeffding- and Bernstein-type inequalities for weighted sums of exchangeable random variables, and [Cheng and Barber \(2026\)](#) extended this perspective to structured weighted sums, including tensor inner products and sequential matrix sums. Other work studies different forms of exchangeable concentration. [Gottschling and Caprio \(2026\)](#) derived Hoeffding-type bounds for unweighted sums in a de Finetti setting, while [Lin et al. \(2026\)](#) studied bounded-difference concentration under infinite exchangeability.

The closest connection is [Lee and Kim \(2026\)](#). They reduce the weighted exchangeable problem first to binary vectors with a fixed number of ones and then, among centered coefficient vectors

with a fixed Euclidean norm, to vectors with at most two distinct coordinate values. The resulting random subset sum is a centered hypergeometric random variable. They also characterize the smallest uniform exchangeable constant through an optimization over these hypergeometric MGFs. We use these two reduction steps without reproving them. The new step required here is to evaluate the remaining hypergeometric optimization exactly.

1.3 Organization of the Paper

Section 2 states the main results and the hypergeometric problem underlying them. Section 3 connects the weighted finite-population statistic to the reduction of Lee and Kim (2026). Sections 4 and 5 prove the hypergeometric MGF bound and determine its smallest uniform constant. Section 6 discusses the scope of the results and the remaining problem at a fixed sample size.

2 Main Results

We now give a formal statement of the weighted result that was outlined in Section 1. We first identify how the coefficient vector enters the variance of an arbitrary signed weighted statistic. The main theorem shows that the same dependence on the coefficients remains in a quadratic MGF bound valid for every $t \in \mathbb{R}$. We then specialize to equal weights to obtain the refinement of Serfling’s inequality, derive the corresponding result for exchangeable variables, and formulate the centered hypergeometric optimization that drives both results.

2.1 Weighted sampling without replacement

Let $w = (w_1, \dots, w_n) \in \mathbb{R}^n$ be a deterministic coefficient vector and define

$$T_w = \sum_{i=1}^n w_i (X_{\pi(i)} - \bar{X}_N), \quad \rho_N(w) = \frac{N \sum_{i=1}^n w_i^2 - (\sum_{i=1}^n w_i)^2}{N-1}.$$

A direct covariance calculation gives $\text{Var}(T_w) = \rho_N(w) \sigma_N^2$. Thus $\rho_N(w)$ determines how the finite-population variance enters the variance of the weighted statistic.

Near $t = 0$, the second-order term of the log MGF is $\text{Var}(T_w)t^2/2$, so its dependence on the coefficient vector is determined by $\rho_N(w)$. We ask whether a quadratic MGF bound valid for every $t \in \mathbb{R}$ can retain this same dependence on w . The following theorem gives such a bound and identifies the smallest multiplier that depends only on N .

Theorem 1. *For every fixed population $X \in [a, b]^N$, every $1 \leq n \leq N$, every deterministic coefficient vector $w \in \mathbb{R}^n$, and every $t \in \mathbb{R}$,*

$$\log \mathbb{E}[e^{tT_w}] \leq \frac{\kappa_N}{8} \rho_N(w) (b-a)^2 t^2, \quad (2)$$

where

$$\kappa_N = \begin{cases} 1, & N \text{ is even,} \\ \frac{2}{N \log((N+1)/(N-1))}, & N \text{ is odd.} \end{cases}$$

Moreover, κ_N is the smallest constant depending only on N for which the MGF inequality (2) holds uniformly over all bounded populations, sample sizes, coefficient vectors, and $t \in \mathbb{R}$.

When N is even, the smallest uniform multiplier is one. When N is odd, it is strictly smaller than one. Section 5 explains why the two cases differ.

2.2 The equal-weight case and Serfling's inequality

We now return to the sample-mean problem that motivated the paper. Taking $w_i = 1/n$ gives

$$T_w = \bar{X}_{S_n} - \bar{X}_N, \quad \rho_N(w) = \frac{N-n}{n(N-1)}, \quad n\rho_N(w) = \frac{N-n}{N-1} = 1 - f_n.$$

Thus, for equal weights, $\rho_N(w)$ reduces exactly to the factor appearing in the sampling variance of the sample mean after multiplication by n . Applying Theorem 1 therefore gives the desired refinement of Serfling's inequality. The following corollary records the resulting MGF and tail bounds.

Corollary 1. *For every $1 \leq n \leq N-1$, every $X \in [a, b]^N$, and every $t \in \mathbb{R}$,*

$$\log \mathbb{E}[\exp\{t(\bar{X}_{S_n} - \bar{X}_N)\}] \leq \kappa_N \frac{N-n}{8n(N-1)}(b-a)^2 t^2. \quad (3)$$

Consequently, for every $u > 0$,

$$\mathbb{P}\{\sqrt{n}(\bar{X}_{S_n} - \bar{X}_N) \geq u\} \leq \exp\left\{-\frac{2u^2}{\kappa_N(1-f_n)(b-a)^2}\right\}. \quad (4)$$

Recall from the finite-population correction in (1) that $f_n = (n-1)/(N-1)$. The same bound holds for the lower tail, and the corresponding two-sided probability is at most twice the right-hand side.

Since $\kappa_N \leq 1$, Corollary 1 improves Serfling's correction to the factor appearing in the exact sampling variance. The three corrections satisfy

$$\underbrace{\kappa_N \frac{N-n}{N-1}}_{\text{Smallest correction uniform over } n} \leq \underbrace{\frac{N-n}{N-1}}_{\text{Factor in the exact sampling variance}} \leq \underbrace{\frac{N-n+1}{N}}_{\text{Serfling's correction}}.$$

The first correction is the smallest one among quadratic MGF bounds that use only the range $[a, b]$ and must hold uniformly over the sample size. Section 5 proves that no smaller correction can hold uniformly over n . For odd N , this uniformity is essential. The result does not imply that the same constant is smallest for every fixed $n > 1$.

2.3 A consequence for weighted exchangeable sums

The finite-population theorem has a second consequence. Exchangeability allows the fixed-population inequality to be transferred to weighted contrasts of bounded exchangeable variables.

For this transfer, we center the coefficient vector after padding it with zeros to length N . Let $\mathbf{1} \in \mathbb{R}^N$ denote the all-ones vector. Define

$$\tilde{w} = (w_1, \dots, w_n, 0, \dots, 0), \quad w^\circ = \tilde{w} - \frac{1}{N}(\mathbf{1}^\top \tilde{w})\mathbf{1}.$$

The entries of w° sum to zero, and a direct calculation gives

$$\rho_N(w) = \frac{N}{N-1} \|w^\circ\|_2^2. \quad (5)$$

Thus only the centered zero-padded coefficients enter $\rho_N(w)$. The same identity will also be used in the reduction to the hypergeometric problem in Section 3.

Now let $X = (X_1, \dots, X_N) \in [a, b]^N$ be exchangeable. For $1 \leq n \leq N$ and $w \in \mathbb{R}^n$, define

$$\mathcal{C}_N^* = \frac{N}{N-1} \kappa_N = \begin{cases} \frac{N}{N-1}, & N \text{ is even,} \\ \frac{2}{(N-1) \log((N+1)/(N-1))}, & N \text{ is odd.} \end{cases}$$

Substituting the centering identity (5) into the finite-population theorem introduces the factor $N/(N-1)$. The following corollary gives the resulting MGF and Chernoff tail bounds.

Corollary 2. *For every $\lambda \in \mathbb{R}$,*

$$\log \mathbb{E} \left[\exp \left\{ \lambda \sum_{i=1}^n w_i (X_i - \bar{X}_N) \right\} \right] \leq \frac{\lambda^2 (b-a)^2}{8} \mathcal{C}_N^* \|w^\circ\|_2^2. \quad (6)$$

The constant $\mathcal{C}_N^$ is the smallest value for which the MGF inequality (6) holds uniformly over bounded exchangeable distributions, coefficient vectors, and λ . If $w^\circ \neq 0$, then for every $u > 0$,*

$$\mathbb{P} \left\{ \sum_{i=1}^n w_i (X_i - \bar{X}_N) \geq u \right\} \leq \exp \left\{ -\frac{2u^2}{(b-a)^2 \mathcal{C}_N^* \|w^\circ\|_2^2} \right\}. \quad (7)$$

If $w^\circ = 0$, the centered contrast is identically zero.

The transfer follows directly from exchangeability. After an independent random permutation, the exchangeable contrast has the same distribution as the corresponding finite-population statistic conditional on the realized vector X . Applying Theorem 1 conditionally and then using the centering identity (5) gives the stated value of $\mathcal{C}_N^*$. The complete argument is given in Appendix A.

This corollary evaluates the smallest exchangeable constant that Lee and Kim (2026) characterized through a centered hypergeometric optimization. For comparison, Barber (2024) obtained the multiplier $(N-1)/(N-H_N) = 1 + O((\log N)/N)$, where $H_N = \sum_{j=1}^N j^{-1}$, and Lee and Kim (2026) improved this to $1 + 3/(2N) + O(N^{-2})$. The exact constant obtained here satisfies

$$\mathcal{C}_N^* = \begin{cases} 1 + N^{-1} + N^{-2} + O(N^{-3}), & N \text{ is even,} \\ 1 + N^{-1} + \frac{2}{3}N^{-2} + O(N^{-3}), & N \text{ is odd,} \end{cases} \quad N(\mathcal{C}_N^* - 1) \rightarrow 1.$$

Thus $\mathcal{C}_N^* = 1 + N^{-1} + O(N^{-2})$, and the coefficient of N^{-1} is exactly one.

2.4 The centered hypergeometric optimization

The finite-population and exchangeable results above both reduce to the same centered hypergeometric MGF. The reduction comes from Lee and Kim (2026) and is written in the present notation in Section 3. We formulate the resulting optimization here because its exact value determines the constants in both applications.

Let $H_{N,K,m} \sim \text{Hypergeometric}(N, K, m)$ and define its centered MGF by

$$G_{N,K,m}(t) = \mathbb{E} \left[\exp \left\{ t \left(H_{N,K,m} - \frac{Km}{N} \right) \right\} \right].$$

The next theorem gives the quadratic bound needed in both applications. It must hold for every K because the reduction can produce a binary population with any number of ones. The factor $m(N - m)/(N - 1)$ keeps the part of the exact hypergeometric variance that depends on the sample size m .

Theorem 2. *For every $N \geq 2$, $0 \leq K \leq N$, $1 \leq m \leq N - 1$, and $t \in \mathbb{R}$,*

$$\log G_{N,K,m}(t) \leq \kappa_N \frac{m(N - m)}{8(N - 1)} t^2. \quad (8)$$

Indeed, the exact variance is

$$\text{Var}(H_{N,K,m}) = \frac{m(N - m)}{N - 1} \frac{K}{N} \left(1 - \frac{K}{N}\right).$$

Thus $m(N - m)/(N - 1)$ is exactly the part of the variance that depends on m . Uniformity in K is essential because the reduction from arbitrary bounded populations produces binary populations with an unrestricted number of ones.

To determine whether the coefficient in Theorem 2 can be reduced, define

$$\kappa_N^* = \max_{1 \leq m \leq N-1} \max_{0 \leq K \leq N} \sup_{t \neq 0} \frac{8(N - 1)}{m(N - m)t^2} \log G_{N,K,m}(t). \quad (9)$$

Thus κ_N^* is the smallest coefficient depending only on N for which the hypergeometric MGF inequality (8) holds uniformly over K , m , and t . The same optimization appears in the characterization of the smallest exchangeable constant in Lee and Kim (2026, Remark 1), with a different normalization,

$$\mathcal{C}_N^* = \frac{N}{N - 1} \kappa_N^*.$$

Lee and Kim (2026) established this characterization but did not evaluate κ_N^* . Theorem 2 gives $\kappa_N^* \leq \kappa_N$. The next proposition supplies the reverse inequality and therefore evaluates the optimization exactly.

Proposition 1. *For every $N \geq 2$,*

$$\kappa_N^* = \kappa_N.$$

If N is even, equality is approached as $t \rightarrow 0$ with $K = N/2$ for every fixed m . If N is odd, equality is attained with $m = 1$, $K = (N - 1)/2$, and $t = 2 \log((N + 1)/(N - 1))$.

The proposition also explains why the smallest constant differs between even and odd N . When N is even, $K = N/2$ is admissible, and letting $t \rightarrow 0$ already forces the multiplier to be at least one. When N is odd, $K = N/2$ is not an integer. The largest ratio is then attained in the Bernoulli case $m = 1$ with $K = (N - 1)/2$ at a nonzero value of t . This identifies the optimum when the bound must hold uniformly over m , but it does not determine the smallest constant for each fixed $m > 1$.

3 Reduction to the Hypergeometric Problem

The proof of Theorem 1 uses the reduction developed by Lee and Kim (2026). Their argument first reduces a bounded population to a binary population and then shows that, for a centered coefficient

vector with fixed Euclidean norm, the largest permutation MGF is attained by a vector with at most two distinct coefficient values. Together, these two steps reduce the weighted problem to a centered hypergeometric MGF.

We do not reprove those steps. Instead, we verify that T_w fits the reduction and record the exact relation between the resulting hypergeometric normalization and $\rho_N(w)$. This makes clear which part of the argument is inherited from [Lee and Kim \(2026\)](#) and which part remains to be proved here.

Write $X_j = a + (b - a)z_j$ with $z_j \in [0, 1]$. Recall that w° is the centered zero padding of w , so $\mathbf{1}^\top w^\circ = 0$. For a fixed $t \in \mathbb{R}$, define $y = (b - a)t w^\circ$. The centering of w° gives

$$tT_w = \sum_{i=1}^N y_i z_{\pi(i)}, \quad \mathbf{1}^\top y = 0.$$

Thus the MGF of T_w is a permutation MGF with a centered coefficient vector y , which is exactly the setting of the reduction in [Lee and Kim \(2026\)](#).

The first step of that reduction allows us to restrict attention to a binary population. Suppose that this population contains K ones. Under a uniform permutation, the locations receiving these ones form a uniformly random K -element subset J_K of $[N]$. Proposition 2 of [Lee and Kim \(2026\)](#) shows that, for a fixed value of $\|y\|_2$, the largest resulting MGF is attained by a coefficient vector taking only two distinct values. Let $A \subset [N]$ be the set on which one value is taken, let $m = |A|$, and let d be the difference between the two values. Then $H_{N,K,m} = |J_K \cap A|$ has the hypergeometric distribution with parameters (N, K, m) , and the centering and norm constraints imply

$$\sum_{i \in J_K} y_i = d \left(H_{N,K,m} - \frac{Km}{N} \right), \quad d^2 = \frac{N}{m(N-m)} \|y\|_2^2.$$

Here m is the number of coordinates of y that take one of the two coefficient values. It is unrelated to the original sample size n in the definition of T_w . This distinction is important because Theorem 2 must hold uniformly over every value of m produced by the reduction.

This representation is the only consequence of the [Lee and Kim \(2026\)](#) reduction needed in the present paper. Because Theorem 2 holds for every K , m , and t , it gives

$$\begin{aligned} \log \mathbb{E}[e^{tT_w}] &\leq \kappa_N \frac{m(N-m)}{8(N-1)} d^2 \\ &= \kappa_N \frac{N}{8(N-1)} \|y\|_2^2 \\ &= \frac{\kappa_N}{8} \rho_N(w) (b-a)^2 t^2, \end{aligned}$$

where the last equality uses $y = (b - a)t w^\circ$ and the centering identity (5). This proves the weighted MGF inequality (2).

The [Lee and Kim \(2026\)](#) reduction therefore identifies exactly what remains to be proved. Their result supplies the reduction to centered hypergeometric MGFs, but it does not determine the smallest quadratic coefficient that works for every K , m , and t . Section 4 proves the required uniform MGF bound using a new recursion, and Section 5 evaluates the smallest multiplier. Appendix A.2 then transfers the corresponding lower bound back to the original weighted statistic.

4 A Uniform Hypergeometric MGF Bound

The previous section reduces the weighted problem to a quadratic bound for the centered hypergeometric MGF that must hold for every K , m , and t . The variance of $H_{N,K,m}$ determines the second-order term of the log MGF at $t = 0$, but the theorem requires control for all t . We obtain this control from an exact recursion for the derivative of the MGF that maps (N, K, m) to $(N - 2, K - 1, m - 1)$.

4.1 An MGF recursion from (N, K, m) to $(N - 2, K - 1, m - 1)$

The following lemma gives the recursion used throughout the argument. Its multiplicative factor is exactly $\text{Var}(H_{N,K,m})$. Because of this exact factor, the induction preserves $m(N - m)/(N - 1)$, which is the dependence on the sample size that appears in the target bound.

Lemma 1. *Let $N \geq 3$, $1 \leq K \leq N - 1$, and $1 \leq m \leq N - 1$, and recall the centered MGF $G_{N,K,m}$ defined in Section 2. For the boundary cases, set $G_{N,K,0}(t) = G_{N,K,N}(t) = 1$. Then*

$$G'_{N,K,m}(t) = 2v_{N,K,m}e^{r_{N,K,m}t} \sinh(t/2)G_{N-2,K-1,m-1}(t), \quad (10)$$

where

$$v_{N,K,m} = \frac{K(N - K)m(N - m)}{N^2(N - 1)} = \text{Var}(H_{N,K,m}),$$

$$r_{N,K,m} = \frac{(N - 2K)(N - 2m)}{2N(N - 2)}.$$

The centered MGF also satisfies the two complement symmetries

$$G_{N,N-K,m}(t) = G_{N,K,m}(-t), \quad (11)$$

$$G_{N,K,N-m}(t) = G_{N,K,m}(-t). \quad (12)$$

Proof. The proof has two ingredients. We first derive a Stein identity for the centered hypergeometric count. We then rewrite the weighted term in that identity as an expectation under a hypergeometric distribution with parameters $(N - 2, K - 1, m - 1)$.

Let $H \sim \text{Hypergeometric}(N, K, m)$ and $\mu = Km/N$. The required Stein identity is

$$\mathbb{E}[(H - \mu)f(H)] = \frac{1}{N}\mathbb{E}[(K - H)(m - H)\{f(H + 1) - f(H)\}]. \quad (13)$$

This identity is closely related to the discrete orthogonal-polynomial formulation of Stein's method (Schoutens, 2001). To verify it directly, write $p(h) = \mathbb{P}(H = h)$. For every h in the support for which both terms are defined, the ratio of consecutive hypergeometric probabilities gives

$$(K - h + 1)(m - h + 1)p(h - 1) = h(N - K - m + h)p(h).$$

After shifting the index on the left-hand side and summing over h , we obtain

$$\mathbb{E}[(K - H)(m - H)f(H + 1)] = \mathbb{E}[H(N - K - m + H)f(H)].$$

Subtracting $\mathbb{E}[(K - H)(m - H)f(H)]$ and using

$$H(N - K - m + H) - (K - H)(m - H) = N(H - \mu)$$

proves the Stein identity (13).

Let $H_- \sim \text{Hypergeometric}(N-2, K-1, m-1)$. The relation between the two hypergeometric probability masses is

$$(K-h)(m-h)\mathbb{P}(H=h) = \frac{K(N-K)m(N-m)}{N(N-1)}\mathbb{P}(H_-=h).$$

Indeed, this follows directly from $(K-h)\binom{K}{h} = K\binom{K-1}{h}$, $(m-h)\binom{N-K}{m-h} = (N-K)\binom{N-K-1}{m-h-1}$, and the ratio of the two normalizing binomial coefficients. Summing against $g(h)$ gives

$$\mathbb{E}[(K-H)(m-H)g(H)] = \frac{K(N-K)m(N-m)}{N(N-1)}\mathbb{E}[g(H_-)]. \quad (14)$$

Combining the Stein identity (13) with the reweighting identity (14) lowers all three parameters from (N, K, m) to $(N-2, K-1, m-1)$.

Apply the Stein identity (13) with $f(h) = e^{t(h-\mu)}$ and then use the reweighting identity (14). Writing $\mu_- = (K-1)(m-1)/(N-2)$, we obtain

$$\begin{aligned} G'_{N,K,m}(t) &= v_{N,K,m}(e^t - 1)e^{(\mu_- - \mu)t}G_{N-2,K-1,m-1}(t) \\ &= 2v_{N,K,m}e^{(1/2 + \mu_- - \mu)t}\sinh(t/2)G_{N-2,K-1,m-1}(t). \end{aligned}$$

A direct calculation gives

$$\frac{1}{2} + \mu_- - \mu = \frac{(N-2K)(N-2m)}{2N(N-2)} = r_{N,K,m},$$

which proves the MGF recursion (10). Replacing the K successes by the $N-K$ failures changes the centered count to its negative and gives the success-complement identity (11). Replacing the sampled subset by its complement gives the sample-complement identity (12). $\square$

The recursion has a direct combinatorial interpretation. The factor $(K-H)(m-H)$ counts pairs consisting of an unselected marked item and a selected unmarked item. Removing such a pair leaves a population of size $N-2$ with $K-1$ marked items and a sample of size $m-1$. The multiplicative coefficient $v_{N,K,m}$ is exactly the hypergeometric variance, so the recursion retains the factor $m(N-m)/(N-1)$ that appears in the target quadratic bound.

4.2 The MGF bound with multiplier one

We first prove the desired quadratic bound with multiplier one. This establishes that the factor from the exact finite-population variance remains valid for every t . The only question left afterward is whether the multiplier one can be reduced uniformly.

Proposition 2. *For every $N \geq 2$, $0 \leq K \leq N$, $1 \leq m \leq N-1$, and every $t \in \mathbb{R}$,*

$$\log G_{N,K,m}(t) \leq \frac{m(N-m)}{8(N-1)}t^2. \quad (15)$$

Proof. We compare $G'_{N,K,m}(t)$ with the derivative of the proposed bound $e^{b_{N,m}t^2}$. The recursion expresses $G'_{N,K,m}(t)$ in terms of the MGF at $(N-2, K-1, m-1)$, which is controlled by the

induction hypothesis. Two elementary exponential bounds then show that the resulting derivative does not exceed the derivative of $e^{b_{N,m}t^2}$.

The sample-complement symmetry in (12) allows us to restrict attention to $1 \leq m \leq N/2$. We prove simultaneously by induction on m that the MGF bound (15) holds for every $N \geq 2m$, every K , and every t . The simultaneous formulation is needed because the recursion maps (N, m) to $(N-2, m-1)$.

For $m = 1$, $H_{N,K,1}$ is a Bernoulli random variable supported on $[0, 1]$, so Hoeffding's lemma gives

$$G_{N,K,1}(t) \leq e^{t^2/8},$$

which is the desired bound. Assume the claim at sample size $m-1$ and define

$$b_{N,m} = \frac{m(N-m)}{8(N-1)}, \quad b_{N-2,m-1} = \frac{(m-1)(N-m-1)}{8(N-3)},$$

$$\Delta = b_{N,m} - b_{N-2,m-1} = \frac{(N-1)^2 - 2m(N-m)}{8(N-1)(N-3)}.$$

The cases $K = 0, N$ are deterministic. By the success-complement symmetry in (11), it is enough to treat $t \geq 0$ for the remaining cases.

For the derivative comparison, set

$$x = \frac{|N-2K|}{N}, \quad c = \frac{N-2m}{2(N-2)}.$$

Then $v_{N,K,m}/b_{N,m} = 2(1-x^2)$ and $r_{N,K,m} \leq cx$. The induction hypothesis and Lemma 1 give, for $t > 0$,

$$G'_{N,K,m}(t) \leq 2v_{N,K,m}e^{r_{N,K,m}t} \sinh(t/2)e^{b_{N-2,m-1}t^2}.$$

We compare this with the derivative $2b_{N,m}te^{b_{N,m}t^2}$ of the target bound. Using $1-x^2 \leq e^{-x^2}$ and $2\sinh(t/2)/t \leq e^{t^2/24}$ gives

$$\frac{G'_{N,K,m}(t)}{2b_{N,m}te^{b_{N,m}t^2}} \leq \exp\left\{-x^2 + cxt + \left(\frac{1}{24} - \Delta\right)t^2\right\}$$

$$\leq \exp\left\{\left(\frac{1}{24} + \frac{c^2}{4} - \Delta\right)t^2\right\},$$

where the second inequality uses $-x^2 + cxt \leq c^2t^2/4$. The remaining coefficient is nonpositive because

$$\Delta - \frac{1}{24} - \frac{c^2}{4} = \frac{N(N-4)(N-2)^2 + 3(N-2m)^2}{48(N-1)(N-3)(N-2)^2} \geq 0.$$

Here $N \geq 2m \geq 4$ in the induction step. Hence $G'_{N,K,m}(t) \leq d\{e^{b_{N,m}t^2}\}/dt$. Both functions equal one at $t = 0$, so integration gives $G_{N,K,m}(t) \leq e^{b_{N,m}t^2}$ for $t \geq 0$. The success-complement symmetry extends the result to negative t , and the sample-complement symmetry covers $m > N/2$. $\square$

Combining Proposition 2 with the reduction in Section 3 proves the weighted MGF inequality with multiplier one for every $X \in [a, b]^N$, every $1 \leq n \leq N$, and every deterministic coefficient vector $w \in \mathbb{R}^n$. In particular, the equal-weight specialization proves that Serfling's correction can be replaced by $(N-n)/(N-1)$. The remaining question is whether the multiplier one is the smallest possible uniformly.

5 The Smallest Uniform Constant

Proposition 2 proves the desired MGF inequality with multiplier one. We now ask whether this multiplier can be reduced while keeping the inequality valid for every K , m , and t .

When N is even, the value $K = N/2$ is admissible, and the second-order expansion of the log MGF at $t = 0$ rules out any multiplier below one. When N is odd, $K = N/2$ is not an integer, so the smallest multiplier must be determined differently. We first use the Bernoulli case $m = 1$ to identify the candidate value and then use the recursion from Lemma 1 to control all larger m . A pointwise derivative comparison works for every K except possibly $K = (N - 1)/2$. For that remaining value, the derivative comparison can cross the target and we instead compare the accumulated derivative difference.

5.1 Why even and odd population sizes differ

We first identify the configurations that determine the candidate smallest constant. Suppose that N is even and take $K = N/2$. The variance and the second-order expansion of the log MGF at $t = 0$ are

$$\begin{aligned}\frac{1}{2} \text{Var}(H_{N,N/2,m}) &= \frac{m(N-m)}{8(N-1)}, \\ \log G_{N,N/2,m}(t) &= \frac{1}{2} \text{Var}(H_{N,N/2,m})t^2 + o(t^2) \quad (t \rightarrow 0).\end{aligned}$$

The ratio that defines κ_N^* in (9) therefore converges to one for every fixed m . Hence the multiplier one in Proposition 2 cannot be reduced when N is even.

When N is odd, $K = N/2$ is not an admissible integer. Throughout the remainder of the proof, let $N \geq 3$ be odd and write

$$L_N = \log \frac{N+1}{N-1} = 2 \operatorname{arctanh}(1/N), \quad s_{N,m} = \frac{m(N-m)}{2N(N-1)L_N}.$$

For odd N , the multiplier in Theorem 1 is $\kappa_N = 2/(NL_N)$. The definition of $s_{N,m}$ is chosen so that the odd- N branch of Theorem 2 becomes

$$\log G_{N,K,m}(t) \leq \frac{s_{N,m}t^2}{2}. \quad (16)$$

The case $m = 1$ identifies the candidate constant. Here $H_{N,K,1}$ is Bernoulli with success probability $p = K/N$. If $B \sim \text{Bernoulli}(p)$, the sharp Bernoulli bound of Schlemm (2016, Theorem 1.1) states that, for $0 < p < 1$ and $p \neq 1/2$,

$$\log \mathbb{E}[e^{t(B-p)}] \leq \frac{1-2p}{4 \log\{(1-p)/p\}} t^2, \quad (17)$$

with equality at the nonzero value $t = 2 \log\{(1-p)/p\}$ when $p < 1/2$.

If $x = |1 - 2p|$, the smallest quadratic coefficient in the Bernoulli MGF inequality (17) is $x/\{4 \operatorname{arctanh}(x)\}$. Since $x \mapsto \operatorname{arctanh}(x)/x$ is increasing on $(0, 1)$, the largest coefficient among the admissible probabilities $p = K/N$ occurs when K is as close as possible to $N/2$. For odd N , these values are $K = (N - 1)/2$ and $K = (N + 1)/2$. By symmetry, it is enough to take $K = (N - 1)/2$, which gives

$$\log G_{N,(N-1)/2,1}(t) \leq \frac{s_{N,1}t^2}{2},$$

with equality at $t = 2L_N$. Thus $m = 1$ and $K = (N - 1)/2$ determine the candidate smallest constant for odd N .

5.2 The odd population upper bound

We now prove the odd- N MGF bound (16) for all m . By the two complement symmetries in (11) and (12), it is enough to consider $t \geq 0$ and $1 \leq m \leq (N - 1)/2$. We prove the claim simultaneously by induction on m over all odd N satisfying $N \geq 2m + 1$. The cases $K = 0, N$ are deterministic, and the base case $m = 1$ was established above.

For the induction step, fix odd $N \geq 5$ and $2 \leq m \leq (N - 1)/2$, and assume that the MGF bound (16) holds for $(N - 2, m - 1)$ uniformly over the corresponding value of K . Define

$$\alpha = \frac{N - 2m}{N - 2}, \quad \gamma = s_{N,m} - s_{N-2,m-1} - \frac{1}{12}. \quad (18)$$

The quantity γ is the difference between the quadratic coefficient in the target bound and the coefficient obtained after applying the recursion together with $2 \sinh(t/2)/t \leq e^{t^2/24}$.

Let $\delta = |1 - 2K/N|$, which measures how far K/N is from $1/2$, and let $F(t) = e^{s_{N,m}t^2/2}$ denote the target quadratic bound. Using the induction hypothesis and Lemma 1, the ratio $\mathcal{R}(t)$ of the resulting derivative upper bound to $F'(t)$ satisfies

$$\log \mathcal{R}(t) \leq -\log \kappa_N + \log(1 - \delta^2) + \frac{\alpha \delta t}{2} - \frac{\gamma t^2}{2}.$$

This inequality shows why the values of K closest to $N/2$ require the most care. As $|N - 2K|$ increases, the factor $1 - \delta^2$ becomes smaller and makes the derivative bound easier to control. When K is close to $N/2$, this reduction is weakest, while the positive linear term can still make the derivative ratio exceed one.

Appendices B.1 and B.2 show that $\mathcal{R}(t) \leq 1$ for every $t \geq 0$ whenever $|N - 2K| \geq 3$. They also handle $K = (N + 1)/2$, where $r_{N,K,m} < 0$. Since the two functions agree at $t = 0$, integrating the derivative comparison proves the target MGF inequality (16) for all of these values of K .

The only remaining case is $K = (N - 1)/2$. Here $r_{N,K,m} = \alpha/(2N) > 0$, so the derivative upper bound can exceed the derivative of the target function on part of the positive half-line. The base case $m = 2$ is verified directly in Appendix B.3, and we now assume $m \geq 3$.

For $K = (N - 1)/2$, define

$$\eta_N = \frac{(N^2 - 1)L_N}{2N}, \quad \ell_N = -\log \eta_N. \quad (19)$$

For $K = (N - 1)/2$, the multiplicative coefficient in the derivative comparison is $\eta_N < 1$, and $\ell_N = -\log \eta_N$ records how far this coefficient lies below one. Explicit bounds for γ and ℓ_N are collected in Appendix B.1.

For the remainder of this case, abbreviate

$$\begin{aligned} S &= s_{N,m}, & B &= s_{N-2,m-1} + \frac{1}{12}, \\ r &= \frac{\alpha}{2N}, & a &= \ell_N. \end{aligned}$$

Then $\gamma = S - B$. The induction hypothesis and the same hyperbolic-sine estimate give

$$G'_{N,K,m}(t) \leq v_{N,K,m} t e^{rt+Bt^2/2}. \quad (20)$$

Relative to $F(t) = e^{St^2/2}$, the ratio of the derivative upper bound in (20) to $F'(t)$ is

$$R(t) = \eta_N e^{rt-\gamma t^2/2} = e^{-a+rt-\gamma t^2/2}.$$

If $R(t) \leq 1$ for every $t \geq 0$, integrating the pointwise derivative comparison proves the desired bound. It remains to consider the case in which R exceeds one. Since $\log R(t) = -a + rt - \gamma t^2/2$ is a concave quadratic, this happens exactly when

$$r^2 > 2\gamma a. \quad (21)$$

Under this condition, $R(t) = 1$ has two positive roots, which we denote by $\tau < \beta$.

Between these roots, the derivative upper bound is larger than $F'(t)$, so the pointwise comparison no longer proves the result. We therefore compare the accumulated derivative difference. Define

$$Q(t) = F(t) - 1 - \int_0^t v_{N,K,m} x e^{rx+Bx^2/2} dx.$$

Then $Q'(t) = F'(t)\{1 - R(t)\}$. Hence Q increases on $(0, \tau)$, decreases on (τ, β) , and increases again on (β, ∞) . Its minimum on $[0, \infty)$ can therefore occur only at 0 or at β . Since $Q(0) = 0$, it remains to show that $Q(\beta) \geq 0$.

For the integrated comparison, define

$$c = \frac{\gamma\beta^2}{2}, \quad d = \frac{S\beta^2}{2}. \quad (22)$$

The identity $R(\beta) = 1$ gives $r\beta = a + c$. After the change of variables $t = \beta z$,

$$Q(\beta) = S\beta^2 \int_0^1 z e^{dz^2} [1 - e^{(1-z)(cz-a)}] dz. \quad (23)$$

The integrand changes sign once at $z = a/c$. The remaining problem is to show that the positive area before the sign change dominates the negative area afterward. Lemma 6 in Appendix B.4 gives a sufficient condition for this comparison, and Lemma 7 in Appendix B.5 verifies that condition whenever the crossing condition (21) holds. Consequently, $Q(\beta) \geq 0$.

It follows that $Q(t) \geq 0$ for every $t \geq 0$. Combining this with the derivative bound (20) gives

$$G_{N,K,m}(t) \leq e^{s_{N,m}t^2/2}$$

for $K = (N - 1)/2$. This completes the induction and proves the odd branch of Theorem 2.

5.3 Sharpness

The upper bound is now established for every N . It remains to verify that it attains the value of the optimization defining κ_N^* in (9).

When N is even, the second-order expansion above gives, for every fixed $1 \leq m \leq N - 1$,

$$\frac{8(N-1)}{m(N-m)t^2} \log G_{N,N/2,m}(t) \rightarrow 1 \quad (t \rightarrow 0).$$

Hence $\kappa_N^* \geq 1 = \kappa_N$.

Now suppose that N is odd. The Bernoulli configuration identified above has $m = 1$, $K = (N - 1)/2$, and $t = 2L_N$. Equality in the Bernoulli MGF inequality (17) gives

$$\log G_{N,(N-1)/2,1}(2L_N) = \frac{L_N}{N}.$$

Substituting this value into the definition of κ_N^* in (9) yields

$$\begin{aligned} \kappa_N^* &\geq \frac{8(N-1)}{(N-1)(2L_N)^2} \frac{L_N}{N} \\ &= \frac{2}{NL_N} = \kappa_N. \end{aligned}$$

Since Theorem 2 gives the reverse inequality $\kappa_N^* \leq \kappa_N$, we conclude that $\kappa_N^* = \kappa_N$ for every $N \geq 2$. This proves Proposition 1.

6 Discussion

Our main result shows that the finite-population correction suggested by the exact sampling variance remains valid at the level of a quadratic MGF bound. This conclusion holds for arbitrary signed weighted statistics, and the equal-weight case resolves the correction question raised by Serfling and later by Greene and Wellner. The same analysis gives a sharp Hoeffding-type bound for weighted contrasts of bounded exchangeable variables. In both settings, the smallest uniform constant is determined by the same optimization over centered hypergeometric MGFs and depends on whether the population size is even or odd.

The statement about the smallest constant is specific to quadratic MGF bounds that use only the range $[a, b]$. It does not imply that the resulting Chernoff bound is the best possible tail bound at every threshold. Exact hypergeometric tails can exploit lattice structure and asymmetry that are lost in a quadratic bound. Bounds that use additional variance or distributional information may also be sharper when such information is available.

One question remains particularly natural. For even N , the smallest constant is approached as $t \rightarrow 0$ for every fixed hypergeometric sample size m . For odd N , the uniform optimum occurs at $m = 1$, and the smallest constant for a fixed $m > 1$ is still unknown. Majorization may provide a useful way to study this problem (Ai et al., 2025). Other directions include nonquadratic MGF bounds, extensions to other sampling designs, and structured exchangeable settings such as tensor- and matrix-valued data (Cheng and Barber, 2026). More broadly, both the finite-population and weighted exchangeable problems lead to the same optimization over centered hypergeometric MGFs.

Disclosure on the use of generative AI

ChatGPT 5.6 was used extensively in the development of the mathematical arguments in this work. The authors provided the model with the research questions, mathematical setting, and high-level directions for possible proof strategies. Based on these inputs, the model was used iteratively to explore proof approaches, develop and refine intermediate arguments, and identify gaps or difficulties in preliminary attempts. Several of the proof ideas and intermediate steps appearing in the manuscript emerged through these interactions.

The final proofs were not generated in a single interaction. Rather, the authors queried ChatGPT 5.6 repeatedly, and the arguments presented in the paper took shape through multiple rounds of exploration, mathematical checking, and revision. The model was also used at later stages to improve the organization and exposition of the proofs and the manuscript. In addition, the authors used Codex to formalize the main proofs in Lean 4. The formalization is available at <https://github.com/statchan1106/sharp-serfling-lean>.

The authors carefully reviewed and revised all AI-assisted material and independently verified the mathematical arguments in the final manuscript. The authors take full responsibility for the contents of the manuscript and for all mathematical claims stated therein.

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A Proofs for the Main Weighted Results

Section 3 recalls the reduction of Lee and Kim (2026) from bounded populations to centered hypergeometric random variables. We do not reprove that reduction here. This appendix contains calculations specific to the results of the present paper. We first verify the variance identity used for the weighted statistic. We then prove that the constant in Theorem 1 cannot be reduced and complete the proofs of the two corollaries.

A.1 Verification of the weighted variance identity

Uniform sampling without replacement gives

$$\begin{aligned}\text{Var}(X_{\pi(i)}) &= \sigma_N^2, \\ \text{Cov}(X_{\pi(i)}, X_{\pi(j)}) &= -\frac{\sigma_N^2}{N-1}, \quad i \neq j.\end{aligned}$$

Substituting these covariance identities into the variance of T_w gives

$$\begin{aligned}\text{Var}(T_w) &= \sigma_N^2 \left\{ \sum_{i=1}^n w_i^2 - \frac{2}{N-1} \sum_{1 \leq i < j \leq n} w_i w_j \right\} \\ &= \sigma_N^2 \left\{ \sum_{i=1}^n w_i^2 - \frac{(\sum_{i=1}^n w_i)^2 - \sum_{i=1}^n w_i^2}{N-1} \right\} \\ &= \rho_N(w) \sigma_N^2.\end{aligned}$$

A.2 Proof of the finite-population theorem

Proof of Theorem 1. The weighted MGF inequality (2) follows from the Lee and Kim (2026) reduction recalled in Section 3 together with the hypergeometric MGF bound in Theorem 2. It remains to prove that κ_N cannot be replaced by a smaller constant uniformly over the population, sample size, coefficient vector, and t .

Fix $1 \leq q \leq N-1$, $0 \leq K \leq N$, and $s \neq 0$. Let $A \subset [N]$ have cardinality q , take $n = N$, and define

$$w_i = \begin{cases} \frac{N-q}{N}, & i \in A, \\ -\frac{q}{N}, & i \notin A. \end{cases}$$

Then $\sum_{i=1}^N w_i = 0$. For a binary population containing K ones and $N-K$ zeros,

$$T_w = H_{N,K,q} - \frac{Kq}{N}, \quad \rho_N(w) = \frac{q(N-q)}{N-1}.$$

Hence any constant C_N replacing κ_N uniformly in the MGF inequality (2) must satisfy

$$\log G_{N,K,q}(s) \leq \frac{C_N}{8} \frac{q(N-q)}{N-1} s^2.$$

Equivalently, $C_N \geq 8(N-1) \log G_{N,K,q}(s) / \{q(N-q)s^2\}$. Taking the supremum over $s \neq 0$ and then the maximum over K and q gives $C_N \geq \kappa_N^*$. Proposition 1 gives $\kappa_N^* = \kappa_N$. Therefore κ_N is the smallest constant for which the MGF inequality (2) holds uniformly. $\square$

A.3 Proofs of the corollaries

Proof of Corollary 1. For $w_i = 1/n$, we have $T_w = \bar{X}_{S_n} - \bar{X}_N$ and $\rho_N(w) = (N - n)/\{n(N - 1)\}$. Theorem 1 therefore gives the equal-weight MGF bound (3).

For the tail bound, let $Z = \sqrt{n}(\bar{X}_{S_n} - \bar{X}_N)$. Then, for every $\lambda > 0$,

$$\log \mathbb{E}[e^{\lambda Z}] \leq \kappa_N \frac{N - n}{8(N - 1)} (b - a)^2 \lambda^2.$$

Chernoff's bound therefore gives

$$\mathbb{P}\{Z \geq u\} \leq \inf_{\lambda > 0} \exp\left\{-\lambda u + \kappa_N \frac{N - n}{8(N - 1)} (b - a)^2 \lambda^2\right\}.$$

Optimizing over λ gives the tail bound (4). The lower-tail bound follows by applying the same argument to $-Z$, and the two-sided bound follows by the union bound.

For completeness, the statement about the smallest constant in Section 2 also follows directly from the hypergeometric representation. For a binary population with K ones, $n(\bar{X}_{S_n} - \bar{X}_N) = H_{N,K,n} - Kn/N$. Hence the normalized quadratic MGF ratio in the equal-weight problem coincides with the ratio defining κ_N^* in (9) with $m = n$. Proposition 1 therefore gives sharpness for every fixed n when N is even, while for odd N the uniform optimum is attained at $n = 1$. $\square$

Proof of Corollary 2. Let π be an independent uniformly random permutation of $[N]$. By exchangeability,

$$\sum_{i=1}^n w_i(X_i - \bar{X}_N) \stackrel{d}{=} \sum_{i=1}^n w_i(X_{\pi(i)} - \bar{X}_N).$$

Conditionally on X , Theorem 1 gives

$$\log \mathbb{E}\left[\exp\left\{\lambda \sum_{i=1}^n w_i(X_{\pi(i)} - \bar{X}_N)\right\} \middle| X\right] \leq \frac{\kappa_N}{8} \rho_N(w) (b - a)^2 \lambda^2.$$

The right-hand side is deterministic, so averaging over X yields the same inequality unconditionally. Using the centering identity $\rho_N(w) = N\|w^\circ\|_2^2/(N - 1)$ proves the exchangeable MGF bound (6) with $\mathcal{C}_N^* = N\kappa_N/(N - 1)$.

To show that the constant cannot be reduced, take an exchangeable vector obtained by uniformly permuting one of the deterministic binary populations used in the sharpness construction above. This realizes the same centered hypergeometric MGF, so Proposition 1 shows that no smaller value of $\mathcal{C}_N^*$ can hold uniformly. Chernoff optimization then gives the exchangeable tail bound (7). $\square$

B Technical Details for the Odd Population Bound

This appendix supplies the technical estimates used in the odd- N argument of Section 5. The main text gives the proof structure, including the Bernoulli base case, the pointwise derivative comparison for most values of K , the integrated comparison for $K = (N - 1)/2$, and the final sharpness argument. Here we verify the required parameter bounds, give the detailed pointwise comparison, treat $K = (N - 1)/2$ and $m = 2$ directly, prove the signed integral lemma used when the pointwise comparison fails, and verify its assumptions for the parameters produced by the hypergeometric recursion. The polynomial calculations used to prove the lower bounds for γ are recorded separately in Appendix C.

B.1 Auxiliary parameter bounds

We use the odd- N constant κ_N from Theorem 1, α and γ from their definition (18), and η_N and ℓ_N from their definition (19).

We first record a simple quantitative approximation to the odd- N constant. The bound is only of order N^{-2} , but this precision is sufficient for the derivative comparisons used throughout the odd population argument.

Lemma 2. *For every odd $N \geq 3$,*

$$1 - \frac{1}{2N^2} \leq \kappa_N \leq 1. \quad (24)$$

Proof. Set $x = 1/N \leq 1/3$ and $z = x^2$. Since $\kappa_N = x / \operatorname{arctanh}(x)$, the upper bound follows from $\operatorname{arctanh}(x) \geq x$. Also,

$$\frac{\operatorname{arctanh}(x)}{x} = 1 + \sum_{j \geq 1} \frac{z^j}{2j+1} \leq 1 + \frac{z}{3(1-z)} \leq \frac{1}{1-z/2}.$$

Taking reciprocals gives

$$\kappa_N \geq 1 - \frac{z}{2} = 1 - \frac{1}{2N^2}.$$

□

The next lemma bounds γ , the difference between the target quadratic coefficient and the coefficient produced by one induction step. The lower bounds are used in the pointwise derivative comparison, while the upper bound is used in the signed integral argument for $K = (N-1)/2$.

Lemma 3. *Let $N \geq 5$ be odd and $2 \leq m \leq (N-1)/2$. Then*

$$\gamma \geq \frac{\alpha^2}{6} > 0. \quad (25)$$

If $m \geq 3$ and $u = m-1$, then

$$\gamma \leq \frac{1}{6}, \quad (26)$$

$$\gamma \geq \frac{\alpha^2}{6} + \frac{u}{12N}. \quad (27)$$

If additionally $N \geq 15$ and $u \leq N/7$, then

$$\gamma \geq \frac{\alpha^2}{6} + \frac{u}{7N}. \quad (28)$$

Proof. Using Lemma 2, the three lower estimates reduce, after multiplication by positive denominators, to polynomial inequalities involving $P_0(N, u)$, $P_{12}(N, u)$, and $P_7(N, u)$, where $u = m-1$. Each polynomial is a concave quadratic in u , so it is enough to check the endpoints of the corresponding interval. After the shifts $N = 5 + x$, $N = 7 + x$, and $N = 15 + x$, respectively, the endpoint polynomials have nonnegative coefficients and positive constant terms. Appendix C records the complete expansions. This proves (25), (27), and (28).

For the upper bound, use $\kappa_N \leq 1$ and $\kappa_{N-2} \geq 1 - 1/\{2(N-2)^2\}$. Direct simplification gives

$$\gamma - \frac{1}{6} \leq \frac{u(-N+u+2)(4N^2-17N+17)}{8(N-3)(N-2)^2(N-1)} \leq 0,$$

because $u \leq (N-3)/2$. $\square$

The final auxiliary quantity describes the multiplicative coefficient in the derivative comparison when $K = (N-1)/2$. How far this coefficient lies below one determines whether the derivative ratio can later exceed one, so we need explicit upper and lower bounds of order N^{-2} .

Lemma 4. *For every odd $N \geq 3$,*

$$\frac{2}{3N^2} \leq \ell_N \leq \frac{3}{4N^2}. \quad (29)$$

Proof. Put $x = 1/N$, $z = x^2$, and write $\eta_N = (1-x^2) \operatorname{arctanh}(x)/x = 1-y$. The power series gives

$$y = \sum_{j \geq 1} \frac{2z^j}{(2j-1)(2j+1)} \geq \frac{2z}{3},$$

$$\ell_N = -\log(1-y) \geq y \geq \frac{2z}{3}.$$

For the upper bound, the coefficients from $j = 2$ onward are at most $2/15$, so $y \leq U(z) := 2z/3 + 2z^2/\{15(1-z)\}$. Since $z \leq 1/9$ and $-\log(1-y) \leq y + y^2/\{2(1-y)\}$, it is enough to verify

$$\frac{3z}{4} - U(z) - \frac{U(z)^2}{2\{1-U(z)\}} = \frac{z(75-520z+685z^2-232z^3)}{60(1-z)(15-25z+8z^2)} \geq 0.$$

The denominator is positive on $0 \leq z \leq 1/9$. Define $p(z) = 75 - 520z + 685z^2 - 232z^3$. On this interval,

$$p'(z) = -2(348z^2 - 685z + 260) < 0, \quad p(1/9) = \frac{18488}{729} > 0.$$

Thus $p(z) > 0$, proving the upper bound. $\square$

B.2 Pointwise derivative comparison for most values of K

We verify the pointwise comparison used in Section 5. Let $N \geq 5$ be odd and $2 \leq m \leq (N-1)/2$, and assume that the MGF bound (16) holds at $(N-2, m-1)$. Recall $\delta = |1 - 2K/N|$ and $\alpha = (N-2m)/(N-2)$, and set $h_N = 1/\kappa_N$ for convenience in this appendix. The recursion in Lemma 1 and the induction hypothesis give

$$G'_{N,K,m}(t) \leq 2v_{N,K,m} e^{r_{N,K,m}t} \sinh(t/2) e^{s_{N-2,m-1}t^2/2}$$

for $t \geq 0$. Let $F(t) = e^{s_{N,m}t^2/2}$. Using

$$\frac{2 \sinh(t/2)}{t} \leq e^{t^2/24}, \quad \frac{v_{N,K,m}}{s_{N,m}} = h_N(1-\delta^2), \quad r_{N,K,m} \leq \frac{\alpha\delta}{2},$$

the ratio $\mathcal{R}(t)$ of the derivative upper bound to $F'(t)$ satisfies

$$\log \mathcal{R}(t) \leq \log h_N + \log(1 - \delta^2) + \frac{\alpha \delta t}{2} - \frac{\gamma t^2}{2}.$$

For $|N - 2K| \geq 3$, use $\log(1 - \delta^2) \leq -\delta^2$ together with the lower bound for γ in (25), and then maximize the resulting quadratic in t . This gives

$$\begin{aligned} \log \mathcal{R}(t) &\leq \log h_N - \delta^2 + \frac{\alpha \delta t}{2} - \frac{\alpha^2 t^2}{12} \\ &\leq \log h_N - \frac{\delta^2}{4}. \end{aligned}$$

Moreover, the bounds on κ_N in (24) and the condition $|N - 2K| \geq 3$ give

$$\log h_N < \frac{9}{4N^2} \leq \frac{\delta^2}{4}.$$

Hence $\mathcal{R}(t) \leq 1$ for every $t \geq 0$. Since $G_{N,K,m}(0) = F(0) = 1$, integration proves the target MGF inequality (16) for every K satisfying $|N - 2K| \geq 3$.

It remains to consider the two admissible values closest to $N/2$, namely $K = (N \pm 1)/2$. For $K = (N + 1)/2$, we have $r_{N,K,m} = -\alpha/(2N) < 0$ and $v_{N,K,m}/s_{N,m} = h_N(1 - 1/N^2) = \eta_N$. Hence

$$\mathcal{R}(t) \leq \eta_N e^{r_{N,K,m}t - \gamma t^2/2} < 1$$

for every $t \geq 0$, because $\eta_N < 1$, $r_{N,K,m} < 0$, and $\gamma > 0$. Thus the case $K = (N + 1)/2$ is also controlled by the pointwise comparison.

For $K = (N - 1)/2$, the sign of the linear term reverses and $r_{N,K,m} = \alpha/(2N) > 0$, so the same pointwise argument need not hold. This is the case treated by the integrated comparison in Section 5.

B.3 The case $K = (N - 1)/2$ with $m = 2$

For $K = (N - 1)/2$, the induction begins with $m = 2$, which is not covered by the recursive comparison used for larger m . In this three-point case the smallest quadratic coefficient can be computed directly, and the following lemma verifies that it lies below the target odd- N bound.

Lemma 5. *Let $N \geq 5$ be odd, $K = (N - 1)/2$, $m = 2$, and $L_{N,2} = \log\{(N + 1)/(N - 3)\}$. Then, for every $t \in \mathbb{R}$,*

$$\log G_{N,(N-1)/2,2}(t) \leq \frac{t^2}{NL_{N,2}}, \tag{30}$$

with equality at $t = L_{N,2}$. Moreover, $2/(NL_{N,2}) \leq s_{N,2}$, so the target MGF inequality (16) holds for this case.

Proof. Let $H \sim \text{Hypergeometric}\left(N, (N - 1)/2, 2\right)$. Its probabilities are

$$\mathbb{P}\{H = 0\} = \frac{N + 1}{4N}, \quad \mathbb{P}\{H = 1\} = \frac{N + 1}{2N}, \quad \mathbb{P}\{H = 2\} = \frac{N - 3}{4N}.$$

To study the smallest quadratic coefficient, define

$$\phi(t) = \log \mathbb{E}[e^{t(H-(N-1)/N)}], \quad D(t) = t\phi'(t) - 2\phi(t).$$

With $z = e^t$, direct differentiation gives

$$\phi'''(t) = -\frac{2z(N+1)\{(N-3)z^2 - (N+1)\}Q_N(z)}{\{(N-3)z^2 + 2(N+1)z + (N+1)\}^3},$$

where $Q_N(z) = (N-3)z^2 + (2N-14)z + (N+1) > 0$. For $N = 5$, $Q_5(z) = 2\{(z-1)^2 + 2\} > 0$, while for $N \geq 7$ all coefficients of Q_N are nonnegative. Hence ϕ''' changes sign exactly once at $t_\star = L_{N,2}/2$.

Since $\phi'''(t) > 0$ for $t < t_\star$ and $\phi'''(t) < 0$ for $t > t_\star$, we have $D''(t) = t\phi'''(t) < 0$ for $t < 0$, $D''(t) > 0$ for $0 < t < t_\star$, and $D''(t) < 0$ for $t > t_\star$. Because $D'(0) = 0$, it follows that $D'(t) > 0$ for $t < 0$, and hence $D(t) < 0$ on $(-\infty, 0)$.

On $(0, \infty)$, the derivative D' first increases and then decreases. Since $H - (N-1)/N$ has finite support with upper endpoint $(N+1)/N$, the exponentially weighted distribution concentrates on this upper endpoint as $t \rightarrow \infty$. Hence $\phi'(t) \rightarrow (N+1)/N$ and $t\phi''(t) \rightarrow 0$. Therefore

$$D'(t) = t\phi''(t) - \phi'(t) \rightarrow -\frac{N+1}{N} < 0.$$

Thus D' has a unique positive zero, so D is strictly increasing and then strictly decreasing on $(0, \infty)$. Direct substitution gives

$$\phi(L_{N,2}) = \frac{L_{N,2}}{N}, \quad \phi'(L_{N,2}) = \frac{2}{N}, \quad D(L_{N,2}) = 0.$$

Since $D(t) > 0$ for all sufficiently small $t > 0$, the strict increase and decrease structure implies

$$\begin{aligned} D(t) &> 0, & 0 < t < L_{N,2}, \\ D(t) &< 0, & t > L_{N,2}. \end{aligned}$$

The ratio $\phi(t)/t^2$ extends continuously to the origin with value $\text{Var}(H)/2$. For $t \neq 0$,

$$\frac{d}{dt} \frac{\phi(t)}{t^2} = \frac{D(t)}{t^3}.$$

Since $D(t) < 0$ for $t < 0$, the ratio $\phi(t)/t^2$ is increasing on $(-\infty, 0)$. On the positive half-line, the sign pattern above shows that it increases on $(0, L_{N,2})$ and decreases on $(L_{N,2}, \infty)$. Hence, with the continuous extension at $t = 0$, $\phi(t)/t^2$ attains its global maximum at $t = L_{N,2}$, which proves the MGF bound (30).

It remains to compare this exact quadratic coefficient with $s_{N,2}$. Since $L_{N,2} = L_{N-2} + L_N$ and $x \mapsto 2x \arctanh(1/x)$ is decreasing on $(1, \infty)$,

$$(N-2)L_{N,2} - 2(N-1)L_N = (N-2)L_{N-2} - NL_N \geq 0.$$

This is equivalent to

$$\frac{2}{NL_{N,2}} \leq s_{N,2}.$$

□

B.4 A Signed Area Comparison

For $K = (N - 1)/2$, the proof reduces to showing that a signed integral remains nonnegative even when the pointwise derivative comparison fails. The following deterministic lemma gives a sufficient condition under which the positive contribution before the sign change dominates the negative contribution afterwards.

Lemma 6. *Let $0 < a < c < d$ and assume $c \leq 2a$. If*

$$(2a - c)(d - c) \geq a^3 + cd^2e^d, \quad (31)$$

then

$$\int_0^1 ze^{dz^2} [1 - e^{(1-z)(cz-a)}] dz \geq 0. \quad (32)$$

Proof. Let I denote the integral in the conclusion (32), and define

$$I_0 = \int_0^1 e^{(d-c)z^2 + (a+c)z-a} dz,$$

Integration by parts gives

$$2(d - c)I = (a + c)I_0 + e^{-a} - 1 - \frac{c}{d}(e^d - 1).$$

Since $e^x \geq 1 + x$,

$$I_0 \geq 1 + \frac{d - c}{3} + \frac{c - a}{2}.$$

We also use the bounds

$$1 - e^{-a} \leq a - \frac{a^2}{2} + \frac{a^3}{6}, \quad \frac{e^d - 1}{d} \leq 1 + \frac{d}{2} + \frac{d^2e^d}{6}.$$

Substitution and simplification yield

$$2(d - c)I \geq \frac{(2a - c)(d - c) - a^3 - cd^2e^d}{6}.$$

The sufficient condition (31) therefore implies $I \geq 0$. □

B.5 Verification for $K = (N - 1)/2$

It remains to verify that the conditions in Lemma 6 are satisfied by the quantities generated by the hypergeometric recursion. The next lemma performs this verification under exactly the crossing condition arising in the main argument.

Lemma 7. *Let $N \geq 7$ be odd, let $3 \leq m \leq (N - 1)/2$, and set $K = (N - 1)/2$. Let S, B, r, a, γ be as defined for this case in Section 5.2, and suppose that $r^2 > 2\gamma a$. With c and d defined in (22),*

$$0 < a < c < d, \quad c < 2a,$$

and

$$(2a - c)(d - c) \geq a^3 + cd^2e^d. \quad (33)$$

Consequently, Lemma 6 applies to the signed integral in (23).

Proof. Write $u = m - 1 \geq 2$. The lower bound for γ in (27), the bounds for ℓ_N in (29), and the crossing condition (21) imply $\alpha^2 > 4u/N$. This forces $u < N/7$. Indeed, set $x = u/N$. If $x \geq 1/7$, then $\alpha < 1 - 2x$ and $(1 - 2x)^2 \leq 4x$, contradicting $\alpha^2 > 4x$. Hence $u < N/7$, and since $u \geq 2$, we necessarily have $N \geq 15$. The stronger bound (28) therefore applies and gives $\alpha^2 > 48u/(7N)$. Applying the same argument with $x = u/N$, now using that $(1 - 2x)^2 \leq (48/7)x$ for $x \in [1/10, 1/2]$, yields $u < N/10$. Since $u \geq 2$ and N is odd, it follows that $N \geq 21$, and in particular $\alpha = 1 - 2u/(N - 2) > 3/4$.

The product of the two positive roots of $R(t) = 1$ is $2a/\gamma$. Since β is the larger root, $\beta^2 > 2a/\gamma$, and hence $c = \gamma\beta^2/2 > a$. Define $z_0 = a/c \in (0, 1)$. Squaring $r\beta = a + c$ and using $c = \gamma\beta^2/2$ gives

$$\frac{2r^2}{\gamma a} = 2 + z_0 + \frac{1}{z_0}.$$

On the other hand, the bounds for ℓ_N in (29) and the stronger lower bound for γ in (28) imply

$$\frac{2r^2}{\gamma a} \leq \frac{9}{2} \frac{7N\alpha^2}{7N\alpha^2 + 6u} < \frac{9}{2}.$$

Thus $z_0 > 1/2$, so $c < 2a$. Moreover,

$$\frac{9}{2} - \left(2 + z_0 + \frac{1}{z_0}\right) = \frac{(2 - z_0)(2z_0 - 1)}{2z_0} \geq \frac{27u}{7N\alpha^2 + 6u}.$$

Because $z_0 \in (1/2, 1)$, we have $2/(2 - z_0) \geq 4/3$. Combining this observation with $u < N/10$ and $\alpha^2 \leq 1$ yields

$$2a - c = a \frac{2z_0 - 1}{z_0} \geq \frac{90u}{19N} a. \quad (34)$$

We next lower-bound $d - c$. Since $d - c = (B/\gamma)c$, the upper bound for γ in (26) and $c > a$ give $d - c \geq 6Ba$. At the same time, the bounds on κ_N in (24) and $10u < N$ imply

$$s_{N-2, m-1} \geq \frac{8}{9} \frac{u(N-2-u)}{4(N-3)} \geq \frac{u}{5}.$$

Since $B \geq s_{N-2, m-1}$, it follows that

$$d - c \geq \frac{6u}{5} a. \quad (35)$$

Combining the lower bounds (34) and (35) gives

$$(2a - c)(d - c) \geq \frac{108}{19} \frac{u^2}{N} a^2. \quad (36)$$

It remains to upper-bound the right-hand side of the sufficient condition in (33). Using $S \leq m/4$, $\gamma \geq \alpha^2/6$, $c < 2a$, and $\alpha > 3/4$, we obtain

$$d = \frac{S}{\gamma} c \leq \frac{3m}{\alpha^2} a < \frac{16m}{3} a < \frac{4m}{N^2} < \frac{4(N/10 + 1)}{N^2} \leq \frac{1}{10}.$$

Since $\log(10/9) > 1/10$, this gives $e^d < 10/9$. Consequently,

$$\begin{aligned} a^3 + cd^2e^d &\leq a^3 \left(1 + \frac{5120}{81}m^2 \right) \\ &\leq \frac{5129}{48} \frac{u^2}{N^2} a^2 \\ &\leq \frac{108}{19} \frac{u^2}{N} a^2. \end{aligned} \tag{37}$$

The second inequality uses $a \leq 3/(4N^2)$, $m/u \leq 3/2$, and $u \geq 2$, while the final inequality follows from $N \geq 21$. The lower bound (36) and upper bound (37) establish the sufficient condition (33). Finally, $B > 0$ implies $d > c$, completing the verification of the hypotheses of Lemma 6. $\square$

C Polynomial Certificates for the Case $K = (N - 1)/2$

This appendix records the algebra used for the three lower bounds on γ in Lemma 3. Let $u = m - 1$ and define

$$\begin{aligned} A_{N,m} &= \frac{m(N-m)}{4(N-1)}, \\ \alpha &= \frac{N-2m}{N-2}, \\ \underline{\gamma} &= \left(1 - \frac{1}{2N^2} \right) A_{N,m} - A_{N-2,m-1} - \frac{1}{12}. \end{aligned}$$

The MGF proof gives $\gamma \geq \underline{\gamma}$, and all denominators below are positive on the indicated parameter ranges. Each identity is an exact polynomial identity; the concavity and endpoint checks complete the inequalities without relying on numerical computation.

C.1 The basic lower bound

For $N \geq 5$ and $1 \leq u \leq (N-3)/2$,

$$\underline{\gamma} - \frac{\alpha^2}{6} = \frac{P_0(N, u)}{24N^2(N-3)(N-2)^2(N-1)},$$

where

$$\begin{aligned} P_0(N, u) &= (-4N^4 + 19N^3 - 21N^2 + 48N - 36)u^2 \\ &\quad + (4N^5 - 27N^4 + 59N^3 - 90N^2 + 132N - 72)u \\ &\quad - 3N^4 + 24N^3 - 69N^2 + 84N - 36. \end{aligned}$$

The coefficient of u^2 is negative for $N \geq 5$ because, after $N = 5 + x$, its negative is $4x^4 + 61x^3 + 336x^2 + 737x + 446$. Thus P_0 is concave in u , and it is enough to inspect

$$\begin{aligned} P_0(N, 1) &= 2(N-3)(2N^4 - 11N^3 + 18N^2 - 36N + 24), \\ P_0\left(N, \frac{N-3}{2}\right) &= \frac{(N-3)(N-1)}{4}(4N^4 - 19N^3 + 9N^2 - 12). \end{aligned}$$

After writing $N = 5 + x$, every coefficient in the two remaining factors is positive. Hence $P_0(N, u) > 0$ throughout the interval.

C.2 The lower bound with $u/(12N)$

For $N \geq 7$ and $2 \leq u \leq (N-3)/2$,

$$\underline{\gamma} - \frac{\alpha^2}{6} - \frac{u}{12N} = \frac{P_{12}(N, u)}{24N^2(N-3)(N-2)^2(N-1)},$$

where

$$\begin{aligned} P_{12}(N, u) = & (-4N^4 + 19N^3 - 21N^2 + 48N - 36)u^2 \\ & + (2N^5 - 11N^4 + 13N^3 - 34N^2 + 108N - 72)u \\ & - 3N^4 + 24N^3 - 69N^2 + 84N - 36. \end{aligned}$$

This polynomial is again concave in u . Its endpoint values are

$$\begin{aligned} P_{12}(N, 2) &= 4N^5 - 41N^4 + 126N^3 - 221N^2 + 492N - 324, \\ P_{12}\left(N, \frac{N-3}{2}\right) &= \frac{(N-3)(N-1)}{4}(9N^3 - 55N^2 + 48N - 12). \end{aligned}$$

After $N = 7 + x$, the unfactored polynomial and the final cubic become

$$\begin{aligned} & 4x^5 + 99x^4 + 938x^3 + 4091x^2 + 7688x + 4296, \\ & 9x^3 + 134x^2 + 601x + 716, \end{aligned}$$

respectively. Both are positive for $x \geq 0$.

C.3 The stronger lower bound with $u/(7N)$

For $N \geq 15$ and $2 \leq u \leq N/7$,

$$\underline{\gamma} - \frac{\alpha^2}{6} - \frac{u}{7N} = \frac{P_7(N, u)}{168N^2(N-3)(N-2)^2(N-1)},$$

where

$$\begin{aligned} P_7(N, u) = & (-28N^4 + 133N^3 - 147N^2 + 336N - 252)u^2 \\ & + (4N^5 + 3N^4 - 139N^3 + 42N^2 + 636N - 504)u \\ & - 21N^4 + 168N^3 - 483N^2 + 588N - 252. \end{aligned}$$

This polynomial is concave in u . Its endpoint values are

$$\begin{aligned} P_7(N, 2) &= 8N^5 - 127N^4 + 422N^3 - 987N^2 + 3204N - 2268, \\ P_7\left(N, \frac{N}{7}\right) &= \frac{1}{7}(22N^5 - 307N^4 + 1266N^3 - 2781N^2 + 3612N - 1764). \end{aligned}$$

After $N = 15 + x$, these become

$$\begin{aligned} & 8x^5 + 473x^4 + 10802x^3 + 116553x^2 + 568944x + 893592, \\ & \frac{1}{7}(22x^5 + 1343x^4 + 32346x^3 + 382239x^2 + 2198982x + 4863816), \end{aligned}$$

which are positive for $x \geq 0$. This completes all three endpoint arguments used in [Appendix B.1](#).